

❏ 1. Introduction to Configuring Services

❏ What is a Service?

A **service** in Linux is a background process (daemon) managed by the init system (like systemd).

- Services are often started at boot.
- They listen for requests (e.g., HTTP, FTP, NFS).
- Managed using commands like `systemctl`, `service`.

❏ Key Commands (systemd-based systems):

```
bash
```

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```
systemctl start <service>    # Start now systemctl stop <service>    # Stop now systemctl restart <service>    # Restart systemctl enable  
<service>    # Enable at boot systemctl disable <service>    # Disable at boot systemctl status <service>    # Check status
```

❏ 2. Setting Up an NFS Server (Network File System)

❏ What is NFS?

NFS allows file sharing between Linux systems over the network.

❏ Server-Side Setup (e.g., /srv/nfs):

1.

Install NFS Server:

bash

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```
sudo apt install nfs-kernel-server    # Debian/Ubuntu sudo yum install nfs-utils    # RHEL/CentOS
```

2.

Create export directory:

```
bash
```

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```
sudo mkdir -p /srv/nfs/shared sudo chown nobody:nogroup /srv/nfs/shared # Permissions
```

3.

Configure /etc/exports:

```
bash
```

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```
/srv/nfs/shared 192.168.1.0/24(rw,sync,no_subtree_check)
```

4.

Export the directories:

```
bash
```

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```
sudo exportfs -a
```

5.

Enable and start the service:

```
bash
```

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```
sudo systemctl enable --now nfs-server
```

☒ Client-Side Setup:

1.

Install NFS client:

```
bash
```

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```
sudo apt install nfs-common    # Debian/Ubuntu sudo yum install nfs-utils    # RHEL/CentOS
```

2.

Mount NFS share:

bash

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```
sudo mount 192.168.1.100:/srv/nfs/shared /mnt
```

3.

Permanent mount:

Add to /etc/fstab:

bash

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```
192.168.1.100:/srv/nfs/shared /mnt nfs defaults 0 0
```

❏ 3. Setting up an FTP Server

❏ What is FTP?

FTP (File Transfer Protocol) allows transfer of files between systems using TCP port 21.

❏ Server-Side Setup:

1.

Install FTP Server:

bash

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```
sudo apt install vsftpd      # Debian/Ubuntu sudo yum install vsftpd      # RHEL/CentOS
```

2.

Main Config File:
`/etc/vsftpd.conf`

📄 Key Settings to Edit:

`conf`

Copy code

```
anonymous_enable=NO local_enable=YES write_enable=YES chroot_local_user=YES
```

3.

Start and Enable Service:

`bash`

Copy code


```
sudo systemctl enable --now vsftpd
```

4.

Firewall Configuration:

```
bash
```

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```
sudo ufw allow 20/tcp sudo ufw allow 21/tcp
```

☒ FTP Clients:

- GUI: FileZilla
-

CLI:

```
bash
```

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```
ftp <server-ip>
```

❌ Secure FTP Alternative:

- Use **SFTP** (part of SSH) for secure file transfer.

```
bash
```

Copy code

sftp user@host